Entanglement Structure of the Angular Momentum Eigenbasis: Energy-Entanglement Decoupling and Basis-Intrinsic Sparsity

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We characterize the entanglement structure of multi-electron atoms in the angular momentum eigenbasis—the basis whose orbitals are products of hydrogenic radial functions and spherical harmonics, labeled by quantum numbers (n, l, m). Four computational results are presented. First, energy-entanglement decoupling: the off-diagonal one-body Hamiltonian (graph Laplacian topology, coupling constant $\kappa = -1/16$) contributes 10–39% of the correlation energy but less than 0.2% of the entanglement entropy; the off-diagonal electron repulsion contributes effectively 100% of the entanglement. Entanglement entropy scales as $S \sim Z^{-2.56}$ across He-like ions. Second, basis-intrinsic sparsity: among orbital bases for systems with spherical symmetry, the angular momentum eigenbasis is the essentially unique basis producing sparse electron repulsion integrals (ERIs) (up to (l, m)-label-preserving rotations; generic angular-mixing rotations fill them). Any generic (angular-mixing) unitary rotation of orbital indices, no matter how small, fills the ERI tensor from 17% to 100% density in a rapid but graded transition at the identity. The nonzero ERI count is Z-independent, confirming that sparsity is purely geometric. Third, entanglement network topology: orbital mutual information maps reveal a hub migration pattern—$1s \to 2s \to 2p$—that tracks the partially filled shell across the first row of the periodic table (for $Z = 7$–9 both the values and the presence of a nonzero valence network are multiplet-member-dependent). Fourth, core-valence decoupling: the core-valence mutual information falls to $\approx 4 \times 10^{-3}$ by $Z = 4$, decays monotonically through $Z = 6$, and reaches the $\approx 10^{-15}$ floating-point noise floor for $Z \geq 7$ — an exact core-closure property of this basis (the $1s$ occupation is exactly 2), so it constrains the basis rather than justifying the composed fiber bundle factorization used in the GeoVac molecular framework. In the composed molecular architecture, per-block entanglement is R-independent, with core/bond entropy ratios of $40\times$ at matched qubit count. These results establish that the angular momentum eigenbasis decouples one-body energy transport from two-body entanglement, and that the Gaunt sparsity of the GeoVac qubit Hamiltonians — whose Pauli count is exactly linear in the qubit count at fixed basis — is a direct consequence of this basis choice. This is Paper 26 in the GeoVac series.

I. INTRODUCTION

Quantum simulation of fermionic systems encodes second-quantized Hamiltonians as weighted sums of Pauli operators. The number of nonzero Pauli terms, the number of qubitwise-commuting (QWC) measurement groups, and the Pauli 1-norm λ are the dominant resource metrics for near-term variational [1] and fault-tolerant [2] quantum algorithms. All three metrics depend on the choice of single-particle basis: a basis that produces sparse two-body matrix elements yields fewer Pauli terms and lower measurement overhead.

The GeoVac framework [5–7] constructs qubit Hamiltonians in the angular momentum eigenbasis, where single-particle orbitals are products of hydrogenic radial functions and spherical harmonics labeled by (n, l, m). In this basis, the Gaunt selection rules from angular momentum conservation zero out the majority of electron repulsion integrals (ERIs) [9]: at $l_{\max} = 2$, the angular ERI density is 8.52%, meaning 91.48% of all possible two-body matrix elements vanish by symmetry before any property of the potential is invoked. (Corrected 2026-08-30: this read 2.76% / 97.24%, which is Paper 22's stricter *pair-diagonal* D_{pd}—valid only for axially-symmetric or m-decoupled bases, not for the Coulomb rule quoted here.) This sparsity produces qubit Hamiltonians whose Pauli count is exactly linear in the qubit count at fixed basis, for composed molecular systems [7], compared to $O(Q^{3.9–4.3})$ for Gaussian orbital bases.

Two questions about this basis remain open. First, is the sparsity fragile or robust? Does an infinitesimal change of basis destroy it, or does it degrade smoothly? Second, what is the entanglement structure of the ground state in this basis? The angular momentum eigenbasis diagonalizes the one-body Hamiltonian (up to a known off-diagonal coupling $\kappa = -1/16$ from the graph Laplacian [6]); does this one-body near-diagonality translate into a simple entanglement pattern?

This paper answers both questions through systematic computation across He-like ions ($Z = 2$–10) and first-row atoms ($Z = 2$–9) in the graph eigenbasis at $n_{\max} = 2$–4. The results establish four structural properties of the angular momentum eigenbasis that are relevant to quantum simulation resource estimation.

Provenance (register tiers). The energy–entanglement decoupling, the $S \sim Z^{-2.56}$ scaling, the $17.1\% \to 100\%$ basis-intrinsic sparsity transition (with a Z-independent nonzero ERI count), the hub-migration pattern, the core–valence factorization thresholds, the $Z^{-0.85}$ decay of the off-diagonal energy fraction (§II), and the composed per-block R-independence with its $\approx 40\times$ core/bond ratio are MEASURED (computed) results; the $Z \geq 7$ core–valence vanishing at $n_{\max} = 2$ is stronger — a derivation from a machine-verified premise (exact core closure). The reported entropy is the von Neumann entropy of the normalized one-body reduced density matrix throughout.

II. THE THREE-LAYER DECOMPOSITION

A. Hamiltonian structure in the graph basis

The full Hamiltonian for an N-electron atom in the angular momentum eigenbasis decomposes as

$$H = H_{\mathrm{diag}} + H_{\mathrm{h1,off}} + H_{\mathrm{Vee,off}}, \tag{1}$$

where H_{diag} contains the diagonal one-body energies ε_{nlm} and the diagonal (Hartree) electron repulsion, $H_{\mathrm{h1,off}}$ contains the off-diagonal one-body matrix elements from the graph Laplacian (with coupling constant $\kappa = -1/16$), and $H_{\mathrm{Vee,off}}$ contains the off-diagonal electron repulsion integrals (exchange and correlation).

These three layers correspond to three physically distinct contributions:

1. **Rational layer** (H_{diag}): diagonal energies and mean-field repulsion. Produces no orbital entanglement—the ground state is a single Slater determinant with zero entanglement entropy.

2. **Topological layer** ($H_{\mathrm{diag}} + H_{\mathrm{h1,off}}$): adds the graph Laplacian inter-shell couplings. These are one-body operators that redistribute amplitude among orbitals but, being separable, generate negligible entanglement.

3. **Correlation layer** ($H_{\mathrm{diag}}+H_{\mathrm{h1,off}}+H_{\mathrm{Vee,off}}$): adds the off-diagonal electron repulsion. This is the irreducible two-body interaction that generates all entanglement in the ground state.

The decomposition is specific to the angular momentum eigenbasis. In a generic orbital basis, the one-body Hamiltonian has dense off-diagonal elements, and the three layers are not separable.

B. Energy-entanglement decoupling

We compute the ground-state energy and the entanglement entropy $S = -\sum_i (n_i/N)\log(n_i/N)$ — the von Neumann entropy of the normalized one-body reduced density matrix (equivalently the natural-orbital occupation entropy, with n_i the 1-RDM eigenvalues and N the electron number) — for each layer by exact diagonalization (full configuration interaction) of He-like ions at $n_{\max} = 4$. This entropy is a genuine correlation measure: it vanishes exactly when the ground state is a single Slater determinant and is nonzero only through the two-body interaction, which is the property used throughout this paper.

TABLE I. Energy-entanglement decoupling in He-like ions at $n_{\max} = 4$. f_E^{h1} and f_E^{Vee}: fractional contribution of each off-diagonal layer to the correlation energy (note: fractions sum to values exceeding 1.0 because the layers are not orthogonal projections of the correlation energy; cross-terms between the layers contribute positively). f_S^{h1} and f_S^{Vee}: fractional contribution to the entanglement entropy. S_{total}: total entanglement entropy (von Neumann entropy of the normalized 1-RDM) in nats.

Z	f_E^{h1}	f_E^{Vee}	f_S^{h1}	f_S^{Vee}	S (nats)
2	0.387	1.290	0.002	0.998	0.0419
3	0.289	1.570	0.000	1.000	0.0120
4	0.228	1.690	0.000	1.000	0.0054
5	0.194	1.770	0.000	1.000	0.0031
10	0.099	1.881	0.000	1.000	0.0007

Table I presents the fractional contribution of each off-diagonal layer to the total correlation energy $E_{\mathrm{corr}} = E_{\mathrm{FCI}} - E_{\mathrm{diag}}$ and to the total entanglement entropy.

The central result is stark: the graph Laplacian topology ($H_{\mathrm{h1,off}}$) contributes 10–39% of the correlation energy but less than 0.2% of the entanglement entropy. The off-diagonal electron repulsion contributes effectively 100% of the entanglement.

The entanglement entropy scales as a power law across the isoelectronic series:

$$S(Z) \sim Z^{-2.56}, \tag{2}$$

obtained from a log-log fit across $Z = 2$–10. This is consistent with the known result that electron correlation becomes perturbatively small at large Z, where the nuclear attraction dominates the electron repulsion and the ground state approaches a single Slater determinant.

The operator-theoretic origin of both the $f_S^{\mathrm{h1}} \to 0$ decoupling and the $Z^{-2.56}$ scaling is established in Paper 27 [10]. The decoupling follows from a free-fermion identity: any non-degenerate one-body ground state on a finite lattice has identically zero single-orbital entropy in its natural-orbital basis, so $f_S^{\mathrm{h1}} \equiv 0$ up to the (exponentially small) projection defect that maps the natural- orbital basis to the angular-momentum eigenbasis. The Z-scaling follows from the Paper 27 universal curve $S_B = A(\tilde{w}_B/\delta_B)^{\gamma}$: across He-like ions, $\tilde{w}_B = \|V_{ee}^{\mathrm{off}(H_1)}\|_F/\|H_1\|_F \sim 1/Z$ (because $V_{ee} \sim Z$ while $H_1 \sim Z^2$) while δ_B is nearly Z-independent, so $S(Z) \sim Z^{-\gamma}$ with $\gamma \approx 2.5$ in the $n_{\max} = 4, Z \in [2, 10]$ window. The measured exponent -2.546 at $n_{\max} = 4$ agrees with Eq. (2) to within 0.6%.

C. Connection to graph validity

The graph off-diagonal coupling $\kappa = -1/16$ is Z-independent, while the diagonal energies scale as Z^2, so a naive estimate of the relative importance of the graph topology is

$$\frac{|\kappa|}{Z^2/n^2} = \frac{1}{16Z^2/n^2}, \tag{3}$$

which predicts a Z^{-2} decay (3.1% at $Z = 2$ for $n = 1$). The measured f_E^{h1} column of Table I decays only as $\sim Z^{-0.85}$ (log–log fit over $Z = 2$–10) and sits 12×–79× *above* the naive estimate across the table.

Corrected 2026-08-29: an earlier version of this subsection asserted that the naive scaling "matches" the f_E^{h1} column; it does not, in either magnitude or exponent. The discrepancy is structural: f_E^{h1} aggregates off-diagonal coupling over the full spectrum, not a single gap, and no quantitative bridge from κ to this column is currently established.

Below the graph validity boundary at $Z_c \approx 1.84$ (Paper 7), the graph off-diagonal coupling is no longer perturbative: $|\kappa|$ exceeds the diagonal energy gap, and the graph-native CI over-binds (violating the variational bound). Whether the slow $Z^{-0.85}$ decay measured here connects quantitatively to that boundary is open: Table I starts at $Z = 2$ and does not probe the sub-Z_c regime.

III. BASIS-INTRINSIC SPARSITY

A. The angular momentum eigenbasis as a sparsity-isolated point

Paper 22 established that the angular ERI density is determined solely by angular quantum numbers and is independent of the radial potential [9]. We now ask a sharper question: within the space of orbital bases for a spherically symmetric system, is the angular momentum eigenbasis an isolated point of sparsity, or is it surrounded by a family of bases with comparable ERI density?

We test this by applying a unitary rotation $U(\theta) = e^{i\theta G}$ to the spatial orbital basis, where G is a random anti-Hermitian generator, and computing the number of nonzero ERIs as a function of the rotation angle θ.

B. Step-function sparsity transition

At $n_{\max} = 2$ (five spatial orbitals: $1s, 2s, 2p_{-1}, 2p_0, 2p_1$), the graph basis produces 107 nonzero ERIs out of $5^4 = 625$ possible four-index combinations, corresponding to an ERI density of 17.1%. Rotating the basis by any nonzero angle fills the ERI tensor: the density rises to 42.7% at $\theta = 10^{-8}$, 70.2% at 10^{-6}, 83.0% at 10^{-4}, and saturates at 100% (625/625) for $\theta \gtrsim 10^{-2}$.

Corrected 2026-08-29. This subsection previously reported $265/625 = 42.4\%$. That count came from an evaluator that omitted the Coulomb selection rule $m_a + m_b = m_c + m_d$, so 158 of the 265 entries (59.6%) were physically zero. The correction *strengthens* the sparsity claim and reverses the basis-size trend reported below; the qualitative content — that the sparse point is isolated and any angular-mixing departure is dense — is unchanged.

The discontinuity is at the identity itself; the approach to saturation is rapid but graded, not a step at a finite θ:

$$D(\theta) = \begin{cases} 0.171, & \theta = 0 \\ 0.427, & \theta = 10^{-8} \\ 0.702, & \theta = 10^{-6} \\ 0.830, & \theta = 10^{-4} \\ 1.000, & \theta \gtrsim 10^{-2}. \end{cases} \tag{4}$$

(Corrected 2026-08-28: this equation previously reported a two-branch form with $D = 0.992$ for $\theta > 10^{-6}$. Neither value reproduces — saturation is complete at 625/625, not 620/625. The qualitative content is unchanged and if anything stronger.)

Generator dependence. The two endpoints — 0.171 at the identity and 1.000 saturation for $\theta \gtrsim 10^{-2}$ — are properties of the basis and are reproduced by every random antisymmetric generator tested. The three intermediate densities are *not*: across five seeds of the same construction the $\theta = 10^{-8}$ value spans 0.38–0.46 and the $\theta = 10^{-4}$ value spans 0.76–0.83 (the quoted spans are sampling-dependent, not generator-variation limits). The values tabulated above are for one representative generator (seed 42) at the 10^{-10} nonzero-entry cutoff used throughout; they should be read as a profile shape, not as basis constants. The two endpoints are cutoff-robust: the identity-point zeros are *exactly* 0.0 and the smallest nonzero entry is 0.0172, so the 107/625 count is independent of any cutoff below that value; saturation is complete (625/625) at every (θ, seed) tested.

This result is not an artifact of the small basis. At $n_{\max} = 4$ (30 spatial orbitals), the graph basis produces 57,700 nonzero entries out of $30^4 = 810{,}000$, a full-tensor density of 7.12%. The density therefore *improves* with basis size, from 17.1% at $n_{\max} = 2$ to 7.12% at $n_{\max} = 4$. Counting instead over canonical unique tuples ($p \leq q$, $r \leq s$) gives 15,293 of 216,225, a density of 7.07% — and the same improvement is visible within that convention alone (18.2% at $n_{\max} = 2$), so the trend is not an artifact of mixing the two counts. Any rotation again fills the tensor to near-complete density.

Corrected 2026-08-29. The retired text reported 318,720 (39.3%) and 79,465 (36.8%) and concluded that the density was "essentially flat . . . does not degrade with basis size." Those counts carried the same missing selection rule as the $n_{\max} = 2$ figure, and at $n_{\max} = 4$ it inflated them by a factor of 5.5. With the rule imposed the conclusion is not merely restored but reversed in the framework's favour: the density falls with basis size (empirically $\sim M^{-0.49}$ over $M = 5, 14, 30$).

C. Z-independence of ERI count

The nonzero ERI count is identical across all Z at fixed $n_{\max}$: 15,293 canonical-unique nonzero ERIs at $n_{\max} = 4$ for $Z = 2, 3, 10$ (and 107 full-tensor at $n_{\max} = 2$ for $Z = 2, 6, 10$). This confirms that the sparsity pattern is purely geometric—determined entirely by angular momentum conservation (Gaunt selection rules)—and independent of the radial wavefunctions. Changing Z rescales the radial functions $R_{nl}(r) \to Z^{3/2} R_{nl}(Zr)$ but does not alter which angular integrals vanish.

D. Implications for Pauli scaling

The Gaunt-driven Pauli sparsity of the GeoVac composed qubit Hamiltonians [7] is a direct consequence of the Gaunt sparsity. The rapid but graded transition (Eq. 4) shows that this scaling is not continuously connected to the $O(Q^{3.9–4.3})$ scaling of Gaussian bases. The angular momentum eigenbasis sits at a distinguished point in the space of orbital bases: an *essentially* unique point where angular momentum conservation is exactly respected and the ERI tensor is sparse—unique up to (l, m)-label-preserving rotations, which leave the sparsity intact. Any *generic* (angular-mixing) departure from it, however small, incurs the dense-ERI penalty.

IV. ENTANGLEMENT NETWORK TOPOLOGY

A. Orbital mutual information

The entanglement network of an N-electron ground state can be characterized by the orbital mutual information [3, 4]

$$I(i, j) = S(\rho_i) + S(\rho_j) - S(\rho_{ij}), \tag{5}$$

where $S(\rho_i)$ is the von Neumann entropy of the reduced density matrix for spatial orbital i, and ρ_{ij} is the two-orbital reduced density matrix. The mutual information $I(i, j)$ quantifies the total (classical and quantum) correlation between orbitals i and j.

We compute $I(i, j)$ from the FCI ground state for first-row atoms in the graph eigenbasis at $n_{\max} = 2$. (Corrected 2026-08-28: the values reported in this section previously came from $n_{\max} = 3$ and $n_{\max} = 4$ runs while the text stated $n_{\max} = 2$ — one of them, $I(2s, 3s)$, referenced an orbital absent from the $n_{\max} = 2$ basis. Every number below is now the $n_{\max} = 2$ value. The migration pattern itself is unchanged, and is cleaner in the corrected data.)

B. Hub migration across the periodic table

The mutual information matrices reveal a hub migration pattern that tracks the partially filled shell:

Helium ($\mathbf{Z = 2}$, 2 electrons): Star topology centered on $1s$. The dominant correlation is $I(1s, 2s) = 0.748$, with all other orbitals connecting through $1s$, which carries 99.0% of the total mutual information. The $1s$ orbital is the sole entanglement hub.

Lithium ($\mathbf{Z = 3}$, 3 electrons): No clean hub shift. The dominant correlation is again $I(1s, 2s) = 0.218$, and $1s$ and $2s$ are *co-hubs* of that single pair, carrying 89.8% and 91.2% of the total mutual information respectively. Lithium is the transition case, not a $2s$-hub system. (Corrected 2026-08-30: this bullet previously reported $I(2s, 3s) = 1.324$ and "the $1s$ orbital drops to 9%". Both were $n_{\max} = 3$ artifacts — $3s$ does not exist in the $n_{\max} = 2$ basis this section uses, as the note above already flagged — and at $n_{\max} = 2$ the $1s$ share is 89.8%, not 9%. The hub migration begins at beryllium, not lithium.)

Beryllium ($\mathbf{Z = 4}$, 4 electrons): Multi-hub topology, and the first genuine hub shift: $1s$ collapses to 0.5% of the total mutual information while $2s$ becomes the primary hub (65.1%), with the strongest single pair already a p–p link, $I(2p_{-1}, 2p_{+1}) = 0.202$.

Nitrogen–Fluorine ($\mathbf{Z = 7–9}$, 7–9 electrons): The hub transitions to the $2p(\pm 1)$ pair, which carries the mutual information: $I(2p_{-1}, 2p_{+1}) = 0.148$ (N), 1.063 (O), 0.311 (F) *for the multiplet member the eigensolver returns*. These ground states are 4-, 7- and 6-fold degenerate, and the value is *member-dependent*: it ranges from 0 on single-determinant members up to a sampled maximum of 0.69 (N), 1.34 (O), 1.32 (F). The quoted values are therefore eigensolver-dependent, not basis constants, and even the qualitative reading "the valence network is alive" is member-dependent.

Retracted 2026-08-29. Earlier versions reported 1.837/1.785/1.644 here and an information-theoretic ceiling of $\ln 8 \approx 2.08$ (N/F) and $\ln 16 \approx 2.77$ (O), "attained exactly by explicit members". Both the values and the ceilings were artifacts of a single-orbital entropy routine that built ρ_i diagonal-only, discarding the spin coherence $\langle\uparrow|\rho_i|\downarrow\rangle$ — nonzero for precisely these sector-mixed multiplet members. Because a diagonal Shannon entropy majorizes the true von Neumann entropy, the routine over-reported s_i (e.g. 0.798 against a true 0.313) and the derived "mutual information" violated subadditivity on half the orbital pairs. No analytic ceiling is asserted in its place; the maxima above are MEASURED.

What *is* degeneracy-robust is the core-valence statement, and it holds as a theorem for all three: every determinant in each ground multiplet carries $1s^2$, so ρ_{1s} is pure for any member and $I_{\mathrm{cv}} = 0$ exactly.

The pattern is:

$$\text{hub: } 1s \xrightarrow{Z=3} \{1s, 2s\} \xrightarrow{Z=4} 2s \xrightarrow{Z=5} 2p, \tag{6}$$

following the shell-filling sequence. *Corrected 2026-08-30:* this equation previously read $1s \to 2s$ at $Z = 3$

and $2s \rightarrow$ frozen at $Z = 4$, both of which the $n_{\max} = 2$ measurements contradict.

The freezing statement holds at every transition *except the first.* Measured shares of the total mutual information: $1s$ carries 99.0% at $Z = 2$ and is still at 89.8% at $Z = 3$ (alongside $2s$ at 91.2%)—at lithium the two are co-hubs of a single dominant pair, and $1s$ is not frozen. It collapses only at $Z = 4$ (0.5%), where $2s$ becomes the hub at 65.1%; thereafter $2s$ itself freezes progressively (19.0% at $Z = 5$, 3.7% at $Z = 6$) as the strongest correlation moves to the $2p$–$2p$ pair ($I = 0.537$ at $Z = 5$, 1.393 at $Z = 6$).

C. Physical interpretation

The hub migration has a simple physical origin: entanglement is generated by the off-diagonal V_{ee} interaction, which is strongest between orbitals that share the same spatial region and are partially occupied. A closed shell ($n_i \approx 2$ for a spatial orbital) has $S(\rho_i) \approx 0$ and contributes negligible mutual information. An open shell ($0 < n_i < 2$) has maximal $S(\rho_i)$ and dominates the entanglement network.

This mechanism is basis-independent in principle, but the angular momentum eigenbasis makes it maximally transparent: the one-body Hamiltonian is nearly diagonal (the graph Laplacian couples only adjacent n-shells with strength $\kappa = -1/16$), so the natural orbital occupations closely track the Aufbau filling order. In a Gaussian basis, where the one-body Hamiltonian has dense off-diagonal elements, the hub structure would be obscured by basis-induced orbital mixing.

V. CORE-VALENCE DECOUPLING

A. Core-valence mutual information

The GeoVac composed geometry [8] factorizes the molecular electronic structure into separate fiber bundles for core and valence electrons, each in its own natural coordinate system. This factorization assumes that the cross-bundle entanglement is negligible. We test this assumption by computing the total core-valence mutual information

$$I_{\text{cv}} = \sum_{i \in \text{core}} \sum_{j \in \text{valence}} I(i, j) \tag{7}$$

for first-row atoms in the graph eigenbasis.

Table II presents the results.

The core-valence mutual information drops sharply with Z, decaying monotonically from 3.7×10^{-3} at beryllium through 1.7×10^{-3} (B) to 6.6×10^{-4} (C), and then reaching the floating-point floor from nitrogen onward. The vanishing at $Z \geq 7$ is not a small number but an *exact* property of this basis: every determinant in the ground multiplet carries $1s^2$, so ρ_{1s} is pure and $I_{\text{cv}} = 0$ identically (measured $\leq 3 \times 10^{-15}$; the $1s$ occupation is exactly 2).

TABLE II. Core-valence mutual information I_{cv} for first-row atoms in the graph eigenbasis ($n_{\max} = 2$). Core orbitals: $1s$. Valence orbitals: $2s, 2p$.

Atom	Z	I_{cv}
He	2	0.751
Li	3	0.228
Be	4	3.7×10^{-3}
B	5	1.7×10^{-3}
C	6	6.6×10^{-4}
N–F	7–9	$< 10^{-14}$ (exact)

This constrains the basis rather than justifying the composed factorization. Exact closure says the $1s$ orbital is *unentangled in this basis at this truncation*, which is a statement about the angular momentum eigenbasis at $n_{\max} = 2$; it is consistent with the composed fiber-bundle ansatz but does not establish it, since the ansatz's content is about the molecular setting where the core and valence blocks carry different effective charges.

Helium ($I_{\text{cv}} = 0.751$) is the exception: with only two electrons, there is no core-valence separation, and the entire system is entangled. Lithium ($I_{\text{cv}} = 0.228$) is intermediate: the $1s$ core is partially decoupled from the $2s$ valence electron, but the separation is not yet clean. This is consistent with the known difficulty of treating lithium in the composed framework (the PK pseudopotential is the accuracy bottleneck [8]).

B. Molecular entanglement: R-independence

In the composed architecture, each orbital block (core, bond pair, lone pair) has its own effective nuclear charge Z_{eff} and its own set of hydrogenic orbitals. The one-body and two-body matrix elements within each block are computed from Z_{eff} alone, with no dependence on the internuclear distance R. The nuclear geometry enters only through the nuclear repulsion energy, which is a classical constant.

This structural feature implies that the per-block entanglement entropy is R-independent. We verify this for the H_2 bond pair in the composed encoding:

$$S_{\text{bond}}(R) = 0.330 \text{ nats} \quad \forall\, R \in [0.5, 10.0] \text{ bohr}. \tag{8}$$

The entropy does *not* approach $\ln 2 \approx 0.693$ nats at large R, which would correspond to the two electrons becoming maximally entangled across the two atomic centers at dissociation. This is a known limitation of the composed architecture: with fixed-exponent blocks, the basis cannot describe bond breaking, where the true wavefunction transitions from a single-determinant bonding state to a multi-reference dissociated state. The composed architecture is designed for single-point energy evaluation at fixed geometries, not for dissociation curves.

C. Core/bond entropy ratio

For LiH in the composed encoding, the core block ($Z_{\text{eff}} = 3$) has $S_{\text{core}} = 0.008$ nats, while the bond block ($Z_{\text{eff}} = 1$) has $S_{\text{bond}} = 0.330$ nats. The ratio is

$$\frac{S_{\text{bond}}}{S_{\text{core}}} \approx 40, \tag{9}$$

consistent with the $S \sim Z^{-2.56}$ scaling law: the core electrons at $Z_{\text{eff}} = 3$ are $3^{2.56} \approx 18$ times less entangled than the bond electrons at $Z_{\text{eff}} = 1$ (the discrepancy from the scaling prediction reflects the different electron counts in core vs. bond blocks).

This ratio quantifies the information-theoretic asymmetry between core and valence: the core contributes negligible entanglement to the total state and can be treated as a classical mean-field background with minimal loss of quantum information.

VI. DISCUSSION

A. Why the angular momentum eigenbasis decouples energy from entanglement

The energy-entanglement decoupling (Table I) has a simple structural origin. The one-body Hamiltonian h_1 is a separable operator: it acts independently on each electron. A separable operator cannot generate entanglement between orbitals. In the angular momentum eigenbasis, h_1 is nearly diagonal (the off-diagonal elements are the graph Laplacian couplings with $\kappa = -1/16$), so $H_{\text{h1,off}}$ is a small perturbation that mixes single-particle states without introducing two-body correlations.

The electron repulsion V_{ee} is the irreducible two-body interaction. It generates entanglement by correlating the occupations of pairs of orbitals. In the angular momentum eigenbasis, the Gaunt selection rules restrict which orbital pairs can interact, producing the sparse entanglement networks observed in Sec. IV.

The decoupling is basis-specific. In a Gaussian orbital basis, the one-body Hamiltonian has dense off-diagonal elements (from the kinetic energy and nuclear attraction integrals in the non-orthogonal basis). These dense one-body couplings mix orbital occupations in ways that blur the distinction between one-body redistribution and two-body correlation. The angular momentum eigenbasis, by nearly diagonalizing the one-body physics, isolates the two-body entanglement in V_{ee} alone.

B. Connection to quantum simulation resources

The energy-entanglement decoupling has a direct implication for quantum simulation. The off-diagonal V_{ee} elements that generate all the entanglement are precisely the elements constrained by the Gaunt selection rules. Fewer nonzero V_{ee} elements means fewer Pauli terms in the Jordan–Wigner encoding, fewer QWC measurement groups, and lower 1-norm—the three resource metrics that dominate near-term quantum algorithm cost.

The composed qubit Hamiltonians [7] exploit this directly: by staying in the angular momentum eigenbasis and respecting the block structure of the composed geometry, the GeoVac Hamiltonians achieve Gaunt-driven Pauli sparsity with 54×–317× fewer Pauli terms than published Gaussian baselines at matched qubit counts (corrected 2026-08-29; the retired pair-diagonal rule gave 51×–1712×).

C. Limitations

Several limitations should be noted.

Basis truncation. The first-row entanglement network results (Sec. IV) use $n_{\max} = 2$ (five spatial orbitals). For $Z \geq 5$, this provides only one valence p-orbital beyond the $1s$ and $2s$ shells. The hub migration pattern is robust across the series, but quantitative mutual information values may shift at larger $n_{\max}$. The He-like ion results (Table I) use $n_{\max} = 4$ (30 spatial orbitals) and are well-converged.

Composed R-independence. The R-independent block entanglement (Eq. 8) is a structural property of the composed architecture, not a physical statement about molecular entanglement. The true molecular ground state has R-dependent entanglement that diverges at dissociation. The composed architecture captures the entanglement within each block accurately but does not describe inter-block entanglement or bond breaking.

d-orbital systems. All results are for s- and p-orbital systems. Whether the energy-entanglement decoupling and the hub migration pattern extend to d-orbital systems (transition metals) is an open question. The Gaunt selection rules apply at all $l_{\max}$ [9], so the basis-intrinsic sparsity is guaranteed, but the entanglement structure may differ.

VII. CONCLUSIONS

Four structural properties of the angular momentum eigenbasis have been established.

First, the one-body Hamiltonian (graph Laplacian topology) contributes significant correlation energy (10–39%) but negligible entanglement ($< 0.2\%$). All entanglement in the ground state originates from the off-diagonal electron repulsion. The entanglement entropy scales as $S \sim Z^{-2.56}$.

Second, among orbital bases for spherically symmetric systems, the angular momentum eigenbasis is an essentially isolated point of ERI sparsity—isolated up to (l, m)-label-preserving rotations. Any *generic* (angular-mixing) unitary rotation of the orbital indices, no matter how small, fills the ERI tensor from 17.1% to 100% density in a rapid but graded transition at the identity.

Third, the orbital mutual information network exhibits a hub migration pattern—$1s \to 2s \to 2p$—that tracks the partially filled shell across the first row. The migration begins at beryllium: helium and lithium are both $1s$-anchored, and lithium is a $1s/2s$ co-hub rather than a $2s$-hub system.

Fourth, core-valence mutual information falls to $\approx 4 \times 10^{-3}$ by $Z = 4$, decays monotonically through $Z = 6$, and vanishes exactly for $Z \geq 7$ — a property of the basis, consistent with but not establishing the composed fiber bundle factorization.

These results establish that the Gaunt sparsity of the GeoVac qubit Hamiltonians is a direct consequence of the angular momentum eigenbasis choice: the basis that decouples one-body energy from two-body entanglement is the same basis that produces sparse ERIs, and both properties arise from the underlying angular momentum conservation symmetry.

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